

When a High Score Is an Illusion: Certifying Genuine versus Repackaged Forecasting Skill

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Abstract—Ranks depend on the observations used for comparison. Reusing those observations can add association between forecast and outcome rank contrasts even when the evaluated forecast and outcome stay fixed. We characterize assignments that preserve association between specified population-rank contrasts, including forecast rank minus baseline rank compared with outcome rank minus baseline rank. Conditional on independent training, whole trajectories are sampled independently from a common law, with unrestricted dependence within each trajectory. The expected score separates into its target and an explicit interaction between map pairs. For a redesign, the learned maps, reference law and coefficient row sums stay fixed. Zero weighted reference overlap for every map pair is necessary and sufficient for preservation uniformly over permitted maps and laws. An unbiased three-trajectory kernel estimates the interaction; independent evaluation and validation provide finite-sample lower bounds. Complete U-statistics estimate the same target directly when all draws can be recombined. Sharp shared-baseline ranges, including ties, tighten both constructions. In a Beijing air-quality archive, interaction accounts for 91.4% to 94.5% of seven learned forecasts’ expected shared scores under the empirical archive law. Separate results address category-fitting error and temporal feedback. In a matched category-control null experiment, distinct references reduce rejections from 402 to 41 out of 1,000 panels.

Index Terms—forecast evaluation, residual covariance, bias correction, rank statistics, multiple testing, panel forecasting

I. INTRODUCTION

A forecast that repeats last week’s sales can be useful. To study its contribution relative to that baseline, we must first specify what the comparison measures [1], [2]. We study forecast rank minus baseline rank compared with outcome rank minus baseline rank.

Ranks count how often a value exceeds reference values, splitting ties equally. Each reference provides forecast, outcome and baseline values. A reference with a lower forecast and outcome but a higher baseline than those being evaluated makes both contrasts positive. Changing the reference can move both contrasts while the evaluated values stay fixed. Which assignments preserve the intended association?

a) *Reference designs that preserve the target*: A redesign keeps the learned maps, reference law and coefficient row sums fixed. We condition on independent training and independent whole trajectories from that law; within-trajectory dependence is unrestricted. The expected score separates into its target and an explicit map-pair interaction. We prove that

zero weighted overlap for every map pair characterizes target preservation uniformly over permitted maps and laws.

b) *From design to a certificate*: An unbiased three-trajectory kernel estimates the interaction. Independent evaluation and validation give finite-sample lower bounds. Complete U-statistics directly estimate the same target when all draws can be recombined. Sharp shared-baseline ranges tighten both constructions. Full baseline adjustment is a separate target; category and temporal branches retain their own conditions (Table I). Figure 1 shows the inputs. Experiments distinguish calibration, certification and subsequent forecast loss.

II. RELATED WORK

Forecast evaluation separates extra information from loss improvement [1], [3]. Clark–West adjusts nested-model squared-loss comparisons [4]; sequential methods bound average predictable score differences [5]. We target fitted rank association under independent trajectory sampling.

Each scalar component of our symmetric kernel is one twelfth of the classical order-three Spearman kernel [6, Example 2.2]. We characterize target-preserving assignments through map-pair interactions. Preprocessing can already induce either bias sign [7]. Aggregate validation specializes higher-order bias estimation [8], [9]: for fixed fits and known category masses, separate validation streams bound the signed error product. McGrath–Mukherjee studies tuning and splitting for optimal conditional-covariance estimation [10].

GCM and cross-fitting require regression-error conditions [11]–[13]; kernel, partial-copula and empirically calibrated tests address conditional independence [14]–[16]. Concentration and betting supply our sampling tools [17]–[19]. Gini covariance and semi-supervised U-statistics use additional forecast information [20], [21]; recursive adjustments precede our explicit temporal source identities [22], [23].

III. METHODOLOGY

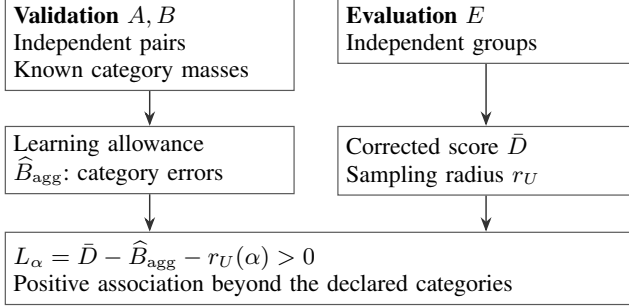
A. Target and interpretation

Let X be a forecast, Y its outcome and Z the declared controls. Panel analyses use standardized within-period average ranks. Let $m_X = \mathbb{E}[X | Z]$, $m_Y = \mathbb{E}[Y | Z]$ and $r_X = X - m_X$, $r_Y = Y - m_Y$. The full-adjustment reference target is

$$\theta_\infty = \mathbb{E}[r_X r_Y], \quad H_0 : \theta_\infty \leq 0 \quad \text{against} \quad H_1 : \theta_\infty > 0. \quad (1)$$

Training fixes learned maps and coefficients before validation and evaluation.

(a) Category certificate



(b) Whole-trajectory design

W_i independent under one law given $\mathcal{H}$;
dependence within each trajectory is unrestricted.

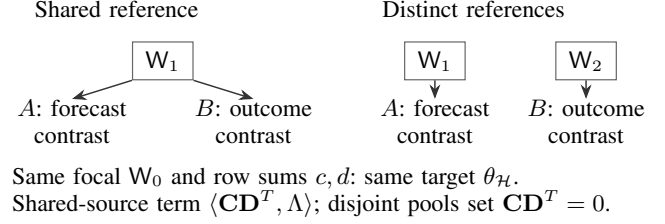


Fig. 1. Independent source designs. (a) Aggregate validation uses exact category masses to bound learning bias; evaluation supplies the score and sampling radius. (b) Whole trajectories share one law and are independent given $\mathcal{H}$. Fixed maps and row sums preserve the fitted target when references are separated. Fresh triples estimate the interaction when sharing is retained.

TABLE I
TARGETS AND REQUIRED INFORMATION.

Target	Required information
$\theta_{\mathcal{H}}$: association between specified fitted rank contrasts	Given $\mathcal{H}$, fixed maps and coefficients; i.i.d. whole trajectories.
ϑ : rank association after category adjustment	Independent categorical peers and validated fits; exact masses for aggregate validation.
θ_{∞} : association after full adjustment for Z	Conditional means or a justified remaining-bias bound, and a valid sampling model.
$\theta_{\mathcal{A}}$: association after additive adjustment	Additive population fits; observed-panel inference needs joint calibration.

Known forecast ranks permit direct kernels; a complete outcome archive permits census. Category-fit training $\mathcal{H}$ and trajectory-map training $\mathcal{H}$ define separate designs.

If $v_X = \mathbb{E}r_X^2 > 0$, the best augmentation $m_Y + tr_X$, $t \geq 0$, reduces squared-error risk by $(\theta_{\infty})_+^2/v_X$: expand its risk as $\mathbb{E}r_Y^2 - 2t\theta_{\infty} + t^2v_X$. For any fitted t and positive θ_{∞} , the gain is $\theta_{\infty}^2/v_X - v_X(t - \theta_{\infty}/v_X)^2$. Certification concerns positive residual association under the stated sampling model. Zero covariance permits nonlinear dependence.

For population fits a_X, a_Y in a declared additive class, conditional centering removes both cross terms in the residual target:

$$\begin{aligned} \theta_{\mathcal{A}} &:= \mathbb{E}[(X - a_X)(Y - a_Y)] \\ &= \theta_{\infty} + \mathbb{E}[(m_X - a_X)(m_Y - a_Y)]. \end{aligned} \quad (2)$$

The second term must be negligible to interpret this as θ_{∞} . Table I separates the targets and required information.

B. Reference design for learned forecasts

For the sales example, let $\phi = (\text{forecast}, \text{baseline})$ and $\psi = (\text{sales}, \text{baseline})$. Row sums c, d fix the fitted contrasts: $(1, -1)$ subtracts the baseline's centered population midrank on each side. This defines $\theta_{\mathcal{H}}$, whereas category adjustment and θ_{∞} remove conditional means. For baseline U uniform on $\{-1, 0, 1\}$, forecast and sales U^2 give $\theta_{\mathcal{H}} = 7/54$ with $c = d = (1, -1)$; full and category-adjusted targets are zero for $Z = U$.

Write $a(v, v') = \mathbf{1}\{v' < v\} + \frac{1}{2}\mathbf{1}\{v' = v\}$ for a midrank comparison. Condition on information $\mathcal{H}$ fixing scalar maps ϕ_s, ψ_t and coefficient matrices $\mathbf{C}, \mathbf{D}$. Whole trajectories $W_0, \dots, W_{J_{\text{ref}}}$ are conditionally i.i.d.; dependence inside each trajectory is unrestricted. Define $p_{sj} = a(\phi_s(W_0), \phi_s(W_j)) - 1/2$ and q_{tj} analogously. Set $A = \sum C_{sj}p_{sj}$, $B = \sum D_{tj}q_{tj}$, $c = \mathbf{C}\mathbf{1}$, $d = \mathbf{D}\mathbf{1}$, $m_s = \mathbb{E}[p_{s1} | W_0, \mathcal{H}]$ and n_t analogously.

Proposition 1 (Conditional trajectory design). *With $\Omega = \mathbf{CD}^T$ and $\Lambda_{st} = \mathbb{E}[\text{Cov}(p_{s1}, q_{t1} | W_0, \mathcal{H}) | \mathcal{H}]$,*

$$\mathbb{E}[AB | \mathcal{H}] = \underbrace{\mathbb{E}[(c^T m)(d^T n) | \mathcal{H}]}_{\theta_{\mathcal{H}}} + \langle \Omega, \Lambda \rangle. \quad (3)$$

For fixed $\mathbf{C}, \mathbf{D}$, preservation for every map and law holds iff $\Omega = 0$ entrywise. On three fresh trajectories, $Q = \Delta p^T \Omega \Delta q / 2$ is unbiased for the interaction, where $\Delta p_s = p_{s1} - p_{s2}$ and $\Delta q_t = q_{t1} - q_{t2}$.

For a fixed map and law, nonzero Ω can still be orthogonal to Λ ; it does not establish bias. For target-preserving redesign, hold the maps, reference law and row sums c, d fixed. Independent training may learn both maps and coefficients. Same-sample adaptation, random rank standardization and dependence across trajectories need separate justification. Forecast maps must also use only information available when issued.

For example, take $\phi = \psi = (U, V)$ with independent fair Bernoulli variables and $\mathbf{C} = \mathbf{D} = (1, -1)^T$. Every map and coefficient row is nonzero; $\Omega = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ has entry sum zero but $\Lambda = I_2/16$ gives interaction $1/8$.

a) *Observable correction and uncertainty:* Use $M \geq 1$ independent evaluation blocks and $n \geq 1$ independent validation triples. Their means are $\bar{S} = M^{-1} \sum_{b=1}^M A_b B_b$ and $\bar{Q} = n^{-1} \sum_{b=1}^n Q_b$. Entrywise norms give valid range-width bounds $W_S = \|\mathbf{C}\|_1 \|\mathbf{D}\|_1 / 2$ and $W_Q = \|\Omega\|_1$. For observations x of size m , set $h(W, m, u) = W \sqrt{\log(1/u)/(2m)}$. For $m \geq 2$, with unbiased sample variance s_x^2 , the preselected hybrid uses

$$\begin{aligned} e(x, W, u) &= s_x \sqrt{\frac{2 \log(2/u)}{m}} + \frac{7W \log(2/u)}{3(m-1)}, \\ r(x, W, u) &= \min\{h(W, m, u/2), e(x, W, u/2)\}, \end{aligned} \quad (4)$$

The pure-range rule sets $r = h$, also used when $m = 1$. For fixed $0 < \delta < \alpha < 1$,

$$L_\alpha = \bar{S} - \bar{Q} - r(S, W_S, \alpha - \delta) - r(Q, W_Q, \delta) \quad (5)$$

has conditional noncoverage at most α by classical concentration [17], [24]. Invert the preselected rule as $p = \inf(\{\alpha \in (\delta, 1) : L_\alpha > 0\} \cup \{1\})$. The released inversion routine uses the range rule; the hybrid is evaluated at a fixed level. A known zero range gives zero uncertainty radius. Total cost is $N_{\text{tot}} = M(J_{\text{ref}} + 1) + 3n$ trajectories, plus training. Columns zero in both matrices can be deleted without changing any score or width. Let $g = 1 + J_{\text{active}}$ count the focal and nonzero reference roles. Minimizing the declared range radius $k_S/\sqrt{M} + k_Q/\sqrt{n}$, $k_S, k_Q > 0$, under $gM + 3n \leq N$ gives $M/n = (3k_S/(gk_Q))^{2/3}$. The constants combine range widths and log-levels, fixed before observing scores.

b) Direct estimation when all trajectories are available: The complete distinct-reference U-statistic uses every draw to estimate the same target. Its six-role kernel has range width at most $W_U = \|c\|_1 \|d\|_1 / 6$. With $q = \lfloor N_{\text{tot}}/3 \rfloor$, the generic-width radius $h(W_U, q, \alpha)$ is less than $h(W_S, M, \alpha - \delta)/\sqrt{6}$ for positive integer M, n, J_{ref} and $W_S > 0$. This compares radii, not realized centers or power. Thus the interaction correction diagnoses an existing design; complete U provides a stronger range bound when the same raw draws can be recombined.

c) Sharper bounds for a shared baseline: For $\phi = (h, b)$, $\psi = (Y, b)$ and $\mathbf{C} = \mathbf{D} = (1, -1)^T$, the sharp ranges, including ties, are $S \in [-1/4, 1]$, $Q \in [-1/2, 2]$ and $U \in [-1/6, 1/3]$ for the six-role symmetric kernel. Substituting widths $5/4$, $5/2$ and $1/2$ tightens both constructions. A declared map identity $h = b$ gives zero kernels and widths; equality on the observed sample alone does not establish that identity.

C. Rank references and learning error

To rank a product, the sales audit compares it with reference products. Consider M independent groups of $N \geq 3$ i.i.d. raw triples (V_i, W_i, Z_i) from a common law, with nuisance fits $f, g \in [0, 1]$ fixed by independent category-fit training H (distinct from the trajectory-map training information $\mathcal{H}$). Using the midrank comparison a , let $a_{ij} = a(V_i, V_j)$ and $b_{ij} = a(W_i, W_j)$. Set $k = N - 1$, $U_{V,i} = k^{-1} \sum_{j \neq i} a_{ij}$ and $U_{W,i} = k^{-1} \sum_{j \neq i} b_{ij}$. Let $F_V(v) = \Pr(V < v) + \frac{1}{2} \Pr(V = v)$ and define F_W analogously. Write $R_V = F_V(V)$, $R_W = F_W(W)$, $m_V = \mathbb{E}[R_V | Z, H]$ and m_W analogously. The target is $\vartheta = \mathbb{E}[(R_V - m_V)(R_W - m_W) | H]$, separate from standardized panel ranks. The shared-reference score $C_i = (U_{V,i} - f(Z_i))(U_{W,i} - g(Z_i))$ becomes

$$D_i = C_i + \frac{U_{V,i}U_{W,i} - k^{-1} \sum_{j \neq i} a_{ij}b_{ij}}{k - 1}. \quad (6)$$

This uses distinct reference indices without discarding observations. Given fitted means, the correction needs only two marginal comparison sums and one joint-product sum per row. Sorting and cumulative counts evaluate all N scores in $O(N \log N)$ time and $O(N)$ working memory, including ties.

Proposition 2 (Reference-reuse decomposition). Write $b_H = \mathbb{E}[(m_V - f)(m_W - g) | H]$ and $\Gamma = \mathbb{E}[a_{12}b_{12} - F_V(V_1)F_W(W_1) | H]$. Then, including ties,

$$\mathbb{E}[C_i | H] = \vartheta + b_H + \Gamma/k, \quad \mathbb{E}[D_i | H] = \vartheta + b_H. \quad (7)$$

Distinct references remove reuse bias; either conditional-mean fit being exact removes b_H . Cauchy-Schwarz gives $|b_H| \leq \varepsilon_f \varepsilon_g$, where $\varepsilon_f^2 = \mathbb{E}[(m_V - f)^2 | H]$ and similarly for g . The remaining fitting error depends on their product.

Set $r_V = R_V - m_V$, $r_W = R_W - m_W$ and, for an independent copy O' ,

$$\begin{aligned} h_V(v) &= \mathbb{E}[(a(V', v) - m_V(Z'))r'_W], \\ h_W(w) &= \mathbb{E}[r'_V(a(W', w) - m_W(Z'))], \\ \psi &= r_V r_W + h_V(V) + h_W(W) - 3\vartheta, \quad \sigma_\psi^2 = \mathbb{E}\psi^2. \end{aligned} \quad (8)$$

Theorem 1 (Covariance boundary and reference reuse). For $M, N \rightarrow \infty$, a fixed common law, exact $f = m_V$, $g = m_W$ and $\sigma_\psi^2 > 0$, let A_j be group j 's mean ($A = C, D$), $\bar{A} = M^{-1} \sum_j A_j$, $\hat{v}_A = (M - 1)^{-1} \sum_j (A_j - \bar{A})^2$ and $T_A = \sqrt{M} \bar{A} / \sqrt{\hat{v}_A}$, set to zero at zero variance. Then

$$\begin{aligned} \sqrt{MN}(\bar{D} - \vartheta) &\Rightarrow \mathcal{N}(0, \sigma_\psi^2), \\ \sqrt{MN}\{\bar{C} - \vartheta - \Gamma/(N - 1)\} &\Rightarrow \mathcal{N}(0, \sigma_\psi^2), \\ N\hat{v}_A &\rightarrow_p \sigma_\psi^2. \end{aligned} \quad (9)$$

At $\vartheta = 0$, $T_D \Rightarrow \mathcal{N}(0, 1)$. At the same boundary, if $M/N \rightarrow \lambda < \infty$, then $T_C \Rightarrow \mathcal{N}(\Gamma\sqrt{\lambda}/\sigma_\psi, 1)$; if $\Gamma > 0$ and $M/N \rightarrow \infty$, its one-sided rejection probability tends to one. At fixed negative or positive ϑ , the corrected rejection probability tends to zero or one, respectively. Under $V \perp W | Z$, $h_V = h_W = 0$ and $\sigma_\psi^2 = \mathbb{E}[r_V^2 r_W^2]$.

The group variance accounts for focal and both reference roles without estimating h_V, h_W . At fixed N with positive group variance, the ordinary group CLT applies.

Corollary 1 (Estimated conditional means). Replace the oracle means in Theorem 1 by fits $f, g \in [0, 1]$ trained on a common sample independent of all evaluation groups. If $\varepsilon_f + \varepsilon_g = o_p(1)$ and $\sqrt{MN} \varepsilon_f \varepsilon_g = o_p(1)$, all its studentized limits remain valid.

a) Learning and sampling allowances: For C_Z categories, m_{val} validation pairs (O, O') are independent of training and evaluation. Focal Z, V, W and reference V', W' supply comparisons with conditional means $m_V(c), m_W(c)$. With n_c focal rows, their Hoeffding intervals have radius $\sqrt{\log(8C_Z/\delta)/(2n_c)}$ and are intersected with $[0, 1]$; empty cells receive $[0, 1]$. Let q_c^+, q_c^- be the largest and smallest of the four endpoint products $(x - f(c))(y - g(c))$ over these two intervals. Obtain binomial upper limits u_c for category probabilities, each with error $\delta/(2C_Z)$ [25]. Define

$$\hat{B}_+ = \max_{p \in \mathcal{P}} \sum_c p_c q_c^+, \quad \hat{B}_- = \min_{p \in \mathcal{P}} \sum_c p_c q_c^-. \quad (10)$$

Here $\mathcal{P} = \{p \geq 0 : \mathbf{1}^T p = 1, p \leq u\}$. Sorted coefficient allocation gives the extrema, including negative terms.

With probability at least $1 - \delta$, $\hat{B}_- \leq b_H \leq \hat{B}_+$. The signed upper allowance can be negative and is no larger than $\hat{B}_{\text{abs}}$ from the largest absolute endpoint errors.

b) Validate the aggregate when category masses are known: The learning bias is one weighted sum. Suppose exact category probabilities p_c are known from control metadata or design; complete forecast or outcome rank metadata are not required. Split independent validation pairs into two streams before inspecting values. In category c , average $a(V, V') - f(c)$ in stream A and $a(W, W') - g(c)$ in stream B , giving $\hat{a}_c, \hat{b}_c$ and counts n_{Ac}, n_{Bc} . The total $m_{\text{val}} = \sum_c (n_{Ac} + n_{Bc})$ pairs cost $2m_{\text{val}}$ raw observations. Their product estimates the bias to subtract (Fig. 1).

Lemma 1 (Aggregate validation allowance). *Condition on training and all focal validation category assignments. For $\mathcal{S} = \{c : n_{Ac}n_{Bc} > 0\}$, let $x = \log(3/\delta)$, $d_c = p_c/(4\sqrt{n_{Ac}n_{Bc}})$ and*

$$\begin{aligned} r_A^2 &= \frac{x}{2} \sum_{c \in \mathcal{S}} \frac{p_c^2 \hat{b}_c^2}{n_{Ac}}, & r_B^2 &= \frac{x}{2} \sum_{c \in \mathcal{S}} \frac{p_c^2 \hat{a}_c^2}{n_{Bc}}, \\ r_{AB} &= \sqrt{2x \sum_{c \in \mathcal{S}} d_c^2} + x \max_{c \in \mathcal{S}} d_c. \end{aligned} \quad (11)$$

Take empty sums and maxima as zero. Let q_c^{full} maximize $(u - f(c))(v - g(c))$ over $u, v \in \{0, 1\}$. Then

$$\hat{B}_{\text{agg}} = \sum_{c \in \mathcal{S}} p_c \hat{a}_c \hat{b}_c + r_A + r_B + r_{AB} + \sum_{c \notin \mathcal{S}} p_c q_c^{\text{full}} \quad (12)$$

satisfies $\Pr(b_H > \hat{B}_{\text{agg}}) \leq \delta$, including ties.

The radii bound the two linear errors and their product, respectively. This is a finite rank-target specialization of classical second-order bias estimation [8], [9] and concentration. It replaces simultaneous mean rectangles by one observable aggregate bound. Exact category masses are essential; plugging in empirical frequencies requires another uncertainty argument. The certificate combines this validation allowance with an independent evaluation-sample bound.

Proposition 3 (Observable reference certificate). *Under the independent common-law reference design, let $\hat{B}$ be an independently validated upper bound on b_H with failure probability δ . Let R be the range of $(a - f(c))(b - g(c))$ over $a, b \in [0, 1]$ and all categories. Fix disjoint triples before inspecting their values. Let s^2 be the sample variance of their six-role symmetric kernels, with denominator $J - 1$, where $J = M \lfloor N/3 \rfloor \geq 2$. For $0 < \delta < \alpha < 1$, write $x = \log\{2/(\alpha - \delta)\}$ and*

$$c_J = \frac{2R}{\sqrt{J(J-1)}} + \frac{R}{3J},$$

$$r_U(\alpha) = \min \left\{ R \sqrt{\frac{x}{2J}}, s \sqrt{\frac{2x}{J}} + c_J x \right\}, \quad (13)$$

$$L_\alpha = \bar{D} - \hat{B} - r_U(\alpha). \quad (14)$$

Then $\Pr(L_\alpha > \vartheta) \leq \alpha$, including ties and zero kernel variance. The mean $\bar{D}$ uses all distinct triples; only its variance estimate uses the fixed disjoint triples.

This combines classical U-statistic concentration and variance bounds [17], [18] with either preselected allowance. Use $\hat{B} = \hat{B}_+$ or $\hat{B}_{\text{agg}}$; taking their random minimum requires a joint error budget. Define $p_{\text{cert}} = \inf(\{\delta < \alpha < 1 : L_\alpha > 0\} \cup \{1\})$. It is super-uniform under $\vartheta \leq 0$. A range-only alternative replaces r_U by $R\sqrt{\log\{1/(\alpha - \delta)\}/(2J)}$ and also allows $J = 1$. Choose the construction before evaluation. For BY at level α_{FDR} , $\delta = \rho\alpha_{\text{FDR}}/(KH_K)$, $0 < \rho < 1$, places the p -value floor below the first family threshold; $H_K = \sum_{j=1}^K 1/j$.

c) Sufficient evaluation budget: Apply Lemma 1 at δ to each sign of stream A , including its corner bounds. The resulting interval $[B_-, B_+]$ has $B_+ = \hat{B}_{\text{agg}}$ and joint coverage at least $1 - 2\delta$. On this event, put $W = B_+ - B_-$. For $\vartheta \geq \vartheta_0 > W$ and $0 < \eta < 1$, the range-only rule has conditional power at least $1 - \eta$ if

$$J > \frac{R^2}{2(\vartheta_0 - W)^2} \left[\sqrt{\log \frac{1}{\alpha - \delta}} + \sqrt{\log \frac{1}{\eta}} \right]^2. \quad (15)$$

Here α is a marginal cutoff; $\alpha = \alpha_{\text{FDR}}/(KH_K)$ suffices for first-threshold BY retention. Cost: $n_{\text{train}} + 2m_{\text{val}} + MN$ raw draws. Choose M, N before fresh evaluation; ϑ_0 is a planning alternative, not an estimated effect. Unconditional power also depends on the random width W .

d) Peer sampling model: The reference correction depends on the sampling design. Condition on a fixed set of N entity attributes, assume that entity pairs are independent across entities and that the two channels are independent within every entity, and let $f_i = \mathbb{E}[U_{V,i}]$ and $g_i = \mathbb{E}[U_{W,i}]$ be the exact entity means. Set $p_{ij} = \mathbb{E}[a_{ij}]$, $q_{ij} = \mathbb{E}[b_{ij}]$; covariance weights each of the k peers equally. Then

$$\mathbb{E}[C_i] = 0, \quad \mathbb{E}[D_i] = -\frac{\text{Cov}_{j \neq i}(p_{ij}, q_{ij})}{k - 1}. \quad (16)$$

Distinct references remove sampling bias for i.i.d. resampled peers but can introduce it for fixed heterogeneous peers. Seeing the same entities repeatedly does not determine which sampling model is appropriate.

D. Feedback through entity effects

Complementary fitting lets earlier outcomes enter evaluated forecasts and evaluated outcomes enter later forecasts. The following identity isolates these routes and the product of fitting errors.

Proposition 4 (Feedback under temporal splitting). *For T observations in contiguous folds with deterministic weights, let $y_r = \alpha + \varepsilon_r$ and $x_r = \sum_{j < r} w_{rj} y_j + \eta_r$. The square-integrable shocks have mean zero and variance $\sigma^2 > 0$; they are mutually uncorrelated and uncorrelated with α and every η_s . Set $w_{sj} = 0$ for $j \geq s$. For evaluation fold I , let E, L be all earlier and later rows, $O = E \cup L$ and $m = |O| > 0$.*

With both entity means fitted on O , set $S_I = \sum_{r \in I} (x_r - \bar{x}_O)(y_r - \bar{y}_O)$. Then

$$\frac{\mathbb{E}S_I}{\sigma^2} = -\frac{\sum_{r \in I, j \in E} w_{rj}}{m} - \frac{\sum_{s \in L, r \in I} w_{sr}}{m} + \frac{|I|}{m^2} \sum_{s, j \in O} w_{sj}. \quad (17)$$

If E, L are nonempty, then $\mathbb{E}[(x_r - \bar{x}_E)(y_r - \bar{y}_L)] = 0$ for every $r \in I$. This latter result also holds for any square-integrable forecast measurable from α , the forecast noises and outcomes before r , provided $\mathbb{E}[\varepsilon_s \mid \alpha, \eta_1, \dots, \eta_T, \varepsilon_1, \dots, \varepsilon_{s-1}] = 0$ for every s .

The first two terms oppose nonnegative forecast weights; the third can reverse their sign. Temporal splitting fits forecast effects earlier and outcome effects later, adapting recursive adjustment and forward orthogonal deviations [22], [23]. This centers raw outcomes with entity means, not the experimental rank/baseline-adjusted panels. The companion preserves the equal-fold closed form and its derivation.

Corollary 2 (Independent-entity inference). *Let S_g be entity g 's mean directional product over fixed eligible rows with nonempty earlier and later folds. Suppose entities, including their forecast noises, are independent and satisfy Proposition 4's centering conditions. If $\sup_g \mathbb{E}|S_g|^4 < \infty$ and $G^{-1} \sum_{g=1}^G \text{Var}(S_g) \rightarrow v > 0$, then $\sqrt{G} \hat{S}/\hat{\text{sd}}(S) \Rightarrow \mathcal{N}(0, 1)$.*

This raw-score test allows fixed observation counts per entity; centering comes from the directional design, without consistent entity-effect fits.

The companion treats explicit lagged-source models; Proposition 1 instead permits general fixed trajectory maps.

E. Observed-panel sensitivity and selection

Panel screens multiply cross-fitted residuals from additive entity effects and baseline bins. Ordered panels use Newey–West variance at the stated Bartlett lag [26]; ratings use cluster variance. Valid inference needs a cluster CLT, consistent variance and negligible fitted-product error [11], [13].

An outcome-free rule abstains on exact copies, matching or reversed cluster orders with ties, and constant ranks. Setting $p^* = 1$ preserves valid inputs because $p^* \geq p$, without repairing invalid ones. Resolution screens combine a fixed bin ladder to cancel a prespecified leading bias term; remaining bias and fitting error must be negligible relative to the standard error. The bias-decay exponents $\beta = 1, 2$ and splines are fixed before evaluation.

Algorithm 1 fixes the rule and family. A valid positive lower bound *certifies* its target at the declared level; *retained* denotes a BY selection, and a guard triggers *abstention*. A raw $p < \alpha$ is a *nominal positive*. Panel outputs remain screens unless their inference conditions hold.

F. Multiple testing and selection

Proposition 5 (Complete-family error control). *For fixed K and super-uniform true-null p -values, BY uses the step-up thresholds $j\alpha_{\text{FDR}}/(KH_K)$, $H_K = \sum_{j=1}^K 1/j$, and controls the false discovery rate (FDR) at α_{FDR} under arbitrary dependence. Fixed K and asymptotically valid inputs give asymptotic control.*

All models remain in the family, including abstentions. Selecting the family or audit rule after viewing results requires a separate conditional-validity argument or independent confirmation.

IV. EXPERIMENTS

A. Which parts of the audit change the evidence?

Figure 2 contrasts resampled and fixed peers: shared scores center for fixed peers; distinct scores depend on effect alignment.

a) *Reference reuse and false positives:* In Fig. 2(b), $Z \sim \text{Bernoulli}(1/2)$ and $V, W \mid Z$ are independent $\text{Bernoulli}(0.2 + 0.6Z)$, giving $\vartheta = 0$ and $\Gamma = 0.0225$. Shared and distinct references reject 402 and 41 of 1,000 paired panels with 8,192 training rows; the corresponding counts with 32 training rows are 677 and 527.

A dependent zero-covariance example rejects 143/3,000 with the full scale and 482/3,000 with the focal-only scale.

b) *Added value of signed learning allowances:* The absolute/signed $\times$ range/variance ablation pairs 100 inspected settings, each with 1,000 repetitions. At 30,976 observations and eight categories with $\vartheta \approx 0.0137$, both range counts are zero; absolute and signed variance bounds reject 274 and 435 times. The paired increase is 16.1 percentage points (pointwise 95% interval [13.1, 18.9]). Figure 3(a) fixes sampling precision across four budgets. All 160 nonpositive method cells record zero rejections.

c) *Comparison with classical sampling methods:* At that eight-category budget, independent-triple and pooled-U variance counts are 433 and 461.

Algorithm 1 Audit a fixed forecast family

- 1: Fix the target, full family and design (Table I) before evaluation.
 - 2: For a certificate, use independent training to fix maps, coefficients, bounds and budgets. For a panel screen, record its fitting protocol.
 - 3: Check source IDs, row sums and forecast availability; justify the branch's sampling assumptions.
 - 4: Return the category bound (14), the trajectory bound (5) or a labeled panel screen.
 - 5: For p -value rules, set abstentions to $p^* = 1$ and leave other p -values unchanged; apply BY for FDR.
 - 6: For fixed-level bounds, test each of K candidates at α/K for Bonferroni familywise control. Return the target and validity status.
-

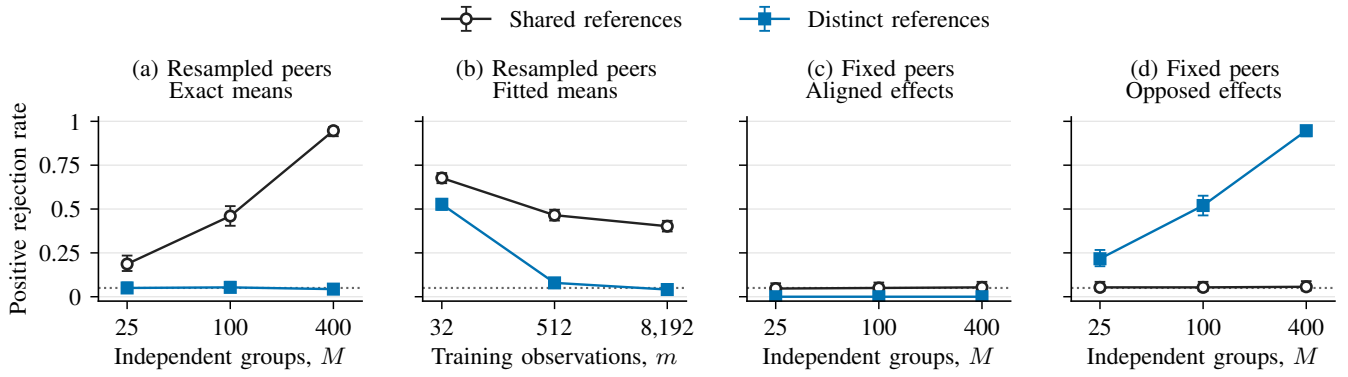


Fig. 2. Reference correction depends on sampling and mean estimation. Positive null rejection rates with pointwise Wilson 95% intervals; the dotted line is 0.05. Panels (a), (c), (d) use exact means, $N = 32$ and 300 replications. Panel (b) uses estimated means, $N = 64$, $M = 400$ and 1,000 replications at each training size. Shared and distinct references use the same evaluation panels. Fixed-peer effects are aligned or opposed between the two channels.

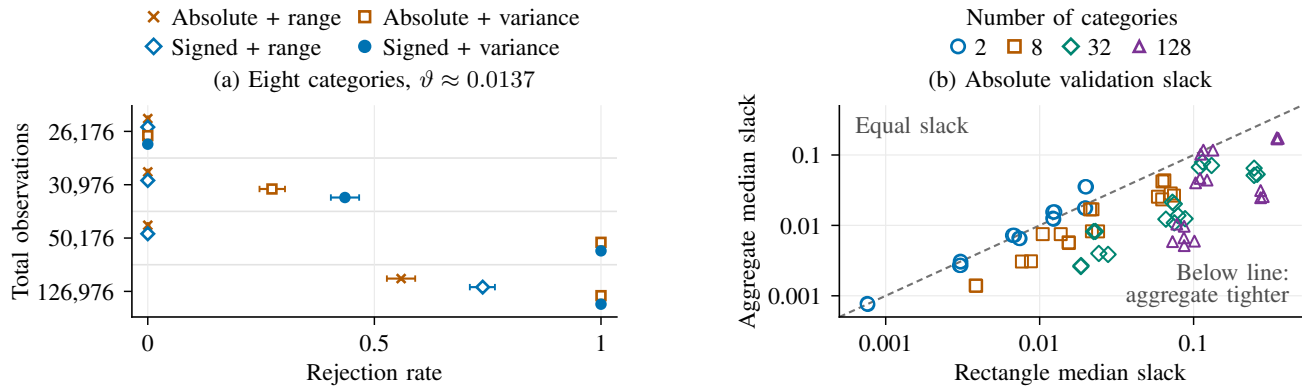


Fig. 3. Learning allowances and sampling precision. (a) Eight-category learned-fit ablation at all four budgets: 1,000 repetitions, pointwise exact binomial 95% intervals, and small vertical method offsets. (b) Absolute validation slack (upper bound minus true bias): 72 settings, 2,000 repetitions each. Aggregation is tighter below equality (62 settings); rectangles above (10). Slack alone does not determine the final association decision.

A separate 300-replication paired comparison at this budget and signed allowance uses a fixed 64-component betting mixture [19], with $p = \min(1, \delta + 1/E)$. Betting detects 296/300 versus 130/300 for the variance rule: a paired gain of 55.3 percentage points (95% interval [47.2, 61.8]). Both null counts are 0/300 (pointwise interval [0, 0.0123]).

d) *Aggregate validation precision*: Figure 3(b) uses 72 settings, 2,000 repetitions and $\delta = 0.05$. With 32 balanced categories, 8,192 validation pairs and exact fits, median bias slack falls from 0.0185 to 0.00260.

Rectangles give smaller median slack in 10/72 settings, including all six two-category cases with opposite fitting errors. All 144 method cells record zero noncoverage. These zero counts do not imply zero risk.

e) *Temporal feedback*: In the raw-score null, gapping reverses complementary feedback from -0.0451 to 0.0655 and rejections from 0 to 458/500. Directional expectation is zero; its worst tested rate is 44/500 (Wilson 95% interval [0.066, 0.116]).

f) *Complete-family error control*: BY tests 10 or 11 raw-score forecasts on 25–400 independent entities with 20 observations: 1,000 families, three positive controls and an

independent 5,000-draw null bank per cell. At 400 entities and 11 models, gapped complementary FDR falls from 0.928 with a Gaussian reference to 0.015 after simulator calibration. For three effects of 0.06, calibrated power is 0.666 for directional blocks and 0.752 for the classical orthogonal moment [23]. Both target the same constant effect; classical weights change the target under heterogeneity. Calibration requires the null bank; the companion’s “Complete-family error and power” table specifies the moments.

Additive-class misspecification gives 300/300 nominal positives, without establishing a zero standardized-rank target. In a near-copy null, linear-plus-spline and 32-bin adjustment reject 46 and 982 of 1,000, respectively, under the same t_{24} and degrees-of-freedom correction.

B. Public forecast protocol

The 41 forecasts use five contiguous folds, additive entity/baseline-bin controls, $\beta = 1, 2$ on $\{8, 12, 16, 24, 32\}$, and splines. Undefined ranks, negligible residuals, standard error $\leq 10^{-10}$ or observed copies receive $p^* = 1$. Directional comparisons fix bins and copy flags; middle support uses folds 2–4. Empty training sides use pooled complementary means.

Applications are Electricity [27] (11 models), M5 [28], MovieLens-25M [29] and Open Source Asset Pricing [30] (10 each). Their baselines are seven-day demand, last-week sales, training item mean and compounded past-year return. Heteroskedasticity and autocorrelation consistent (HAC) lags are 14 daily, 2 weekly and 12 monthly; ratings use user clustering. The companion’s “Directional fitting support by task” table reports fallback and unseen-entity shares, outside the temporal centering result.

V. RESULTS AND ANALYSIS

A. Same forecasts, different screening decisions

On 68,000 fixed middle-fold rows over 17 weeks, retail price/calendar moves from 6.41 to -0.82 . Croston SBA moves from -19.43 to 6.11. Retention gains three models and loses one. Residual-fit cross terms and the product of fit changes decompose the shifts without identifying their causal feedback share.

All-fold complementary fitting retains 10/11 electricity, 7/10 retail, 6/10 ratings and 0/10 asset-pricing models under both exponents and splines. The ratings baseline has mean absolute error (MAE) 0.693 and abstains; alternating least squares has MAE 0.719 and is retained across all four fold/support choices. Ranks, fitting and dependence correction lack joint calibration.

B. From residual evidence to forecast decisions

Electricity boosting trained on 67,392 earlier rows gives $L_{0.05} = 0.0705$ for target 0.0782 in a 70,080-row archive. All seven bounds certify the seasonal forecast. A category-only copy has zero target and lower bound -0.00794 . Four categories cross weekend status with zero/positive seasonal forecasts; 32,768 draws revisit 26,140 existing rows.

Appliances energy [31] and Metro traffic [32] protocols were locally fixed before first download. A midrank repair changed six copy decisions, preserving models and windows. Each task has eight candidates and chronological training, calibration, selection and confirmation. Gates cross 16 training-forecast quantile bins with weekend status, using 32,768 selection-archive draws and eight-candidate BY. Known forecast metadata permit classical covariance kernels [20] and pairs without learned means.

With training-scale clipping (Table II), known-forecast U, pairs and ungated selection choose ExtraTrees: MSE improves 16.95% and 73.87% over Ridge, with zero gain from gating. Reference allowances exceed the associations, causing Ridge fallback.

a) Development diagnosis with category metadata: On four inspected eight-candidate families with matched metadata, fits and 32,768 draws, aggregate and fully averaged second-order allowances both retain 13 with betting and 9 with pooled U. Rectangles with betting retain 3.

b) Chronological news confirmation: News Popularity [33] fixed six candidates before source access. On 1,110 articles, five of nine gates select ungated boosting augmentation; four return Ridge. The augmentation’s clipped loss falls

TABLE II
CONFIRMATION MEAN SQUARED ERROR.

Rule	MSE ($\times 10^{-3}$)	
	Energy	Traffic
Ridge baseline	20.80	5.252
Reference U / betting	20.80	5.252
Ungated augmentation	17.28	1.372
Known-forecast U / pairs	17.28	1.372
Direct-loss bound	20.80	1.372
Convex combination	17.28	1.404

Confirmation MSE after training-scale clipping; lower is better. Primary Ridge baseline, 1,008 energy and 8,713 traffic hours. Reference rules equal Ridge; known-forecast rules equal ungated predictions. Reference rules use the original rectangular validation protocol.

TABLE III
BOOSTING 31: CERTIFICATE DECOMPOSITION.

Quantity (10^{-3})	Original	Reallocated	Triple	Pooled
	shared	shared	U	U
Score mean	8.530	10.355	0.870	0.847
Estimated interaction	8.148	8.477	0.000	0.000
Evaluation radius	4.041	4.930	0.653	0.601
Validation radius	7.886	4.660	0.000	0.000
Lower bound	-11.545	-7.712	0.217	0.246

All methods use 73,728 draws and the shared-baseline ranges. Exact target: 0.852; expected shared score: 9.909. Candidate level 0.05/8; shared validation budget 0.05/16. Shared and triple rules use the hybrid; pooled U uses its variance bound. Lower bound subtracts the interaction and both radii from the mean; calculations use unrounded values.

0.158% and raw MSE rises 12.82% versus Ridge. No gate improves ungated clipped loss.

c) Learned trajectories and actual decisions: Beijing PM2.5 [34] forecasts issue at 14:00 for 15:00. Trajectories contain the target, outcome lags 1, 2, 3, 6, 12, 24 hours (lag 1 is the 14:00 outcome), issue-hour weather/wind and calendar fields. Issue-time measurements are assumed available. Eight models train on 23,006 hours in 2010–2012; selection uses 342 daily trajectories in 2013 and evaluation 350 in 2014. For forecast h and persistence b , $\phi = (h, b)$, $\psi = (Y, b)$ and $\mathbf{C} = \mathbf{D} = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}$. With-replacement draws from these 342 trajectories implement Proposition 1 for the empirical archive law: $M = 16,384$, $J_{\text{ref}} = 2$, $n = 8,192$ with Bonferroni level 0.05/8. The 73,728 draws revisit 2,069 hourly rows, with no new labels. In this inspected-data comparison, shared-baseline ranges tighten all four methods (Table III). For Boosting 31, the exact interaction is 0.00906 of the 0.00991 expected shared score (91.4%). Removing the unused column frees 16,384 draw positions. Reallocation within 73,728 draws gives $M = 13,086$, $n = 15,852$. At that allocation, the sharper ranges reduce its hybrid radius from 0.0121 to 0.00959 (20.9%), with unchanged center.

Neither shared-reference rule certifies a candidate. Triple hybrid certifies 3/8; pooled variance certifies 4/8, adding Boosting 7 ($L = 1.13 \times 10^{-5}$). Selected forecasts are unchanged. Pure-range rules certify none. Census takes 0.010 seconds versus pooled U 0.438 for Boosting 31.

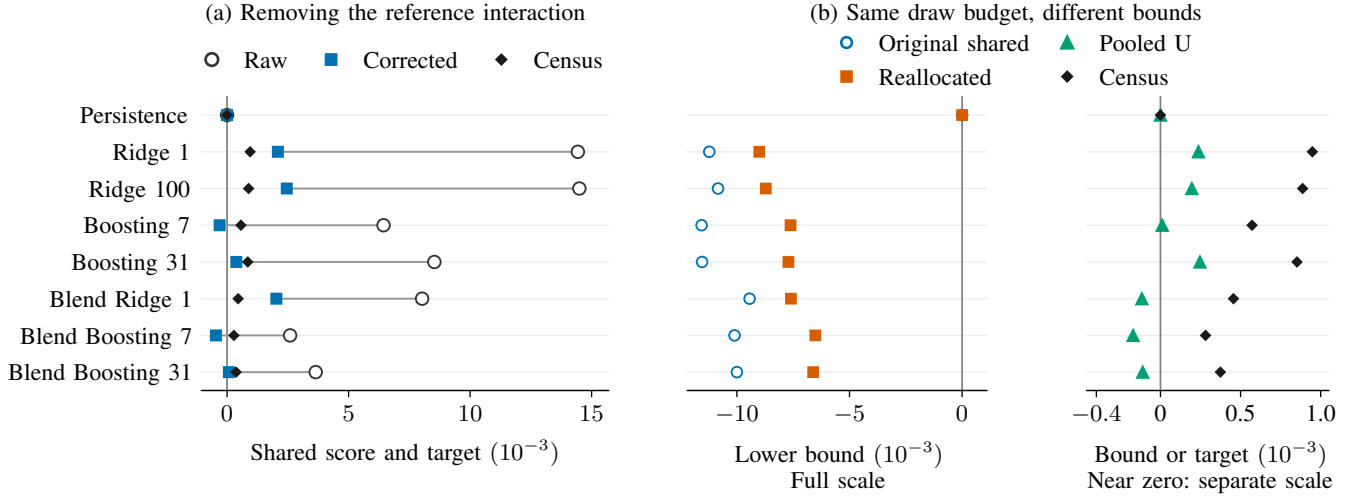


Fig. 4. Reference correction on the Beijing archive with shared-baseline ranges. Left: original raw score, corrected center and exact archive target. Right: hybrid shared bounds and the pooled variance bound; diamonds are exact targets. Original $(M, n) = (16,384, 8,192)$ and reallocated $(13,086, 15,852)$ designs each use 73,728 draws. Ridge numbers are penalty strengths; boosting numbers are maximum leaf counts. Only positive lower bounds certify candidates; Table III also reports the direct-triple bound.

Shared-reference correction selects persistence (MSE 288.32); triple U, pooled U and ungated selection choose boosting (MSE 295.94). The loss gain is 7.62 (2.57%), with subsequent descriptive paired four-week bootstrap interval $[-36.48, 51.38]$ in MSE units. Gating adds zero gain over choosing persistence directly. An origin/target-hour mismatch was repaired after inspecting outcomes. Both cohorts and their results are released.

d) *Prespecified finite-law checks*: Two pre-simulation protocols test seven IID laws and two budgets, with four candidates per family: the baseline, a forecast, its reversal and a constant. Three methods each use two bounds, giving $7 \times 2 \times 3 \times 2 = 84$ cells with 300 repetitions each. A nonidentity forecast has zero target and interaction $1/32$. All cells have 0/300 family noncoverage and false certifications (pointwise two-sided 95% interval $[0, 0.0122]$). For target $1/135$ at 12,288 draws, both U hybrid rules certify 300/300 versus shared correction's 0/300; at 3,072 draws all three certify 0/300. The companion gives all results under common structural ranges.

VI. CONCLUSION

Our identities locate induced association in rank-based forecast comparisons. For fixed maps, reference laws and row sums, separate references or estimate their interaction. When all trajectories are available, complete U-statistics give a stronger range bound for the same target.

a) *Limitations*: Observed-panel ranks, additive fits and dependence corrections lack joint calibration. Aggregate validation needs exact masses and independent pairs; trajectory bounds need conditionally i.i.d. blocks and concern a fitted rank target. Temporal results require their stated conditions; simulator calibration needs a null law. Archives establish neither future-loss nor acquisition gains; chronological studies do not establish a reproducible incremental gating benefit. Earlier-task comparisons are developmental. Restricted training and

historical availability remain qualified. Non-retention does not prove no predictive value.

b) *Code and data availability*: Code, licensed public data, results and derivations are available.¹ The restricted application releases only model-level and simulation aggregates; observations, identifiers, predictions, residuals and loaders remain private.

APPENDIX PROOFS

Proof of Proposition 1. Conditional independence gives

$$\mathbb{E}[p_{sj}q_{tk} \mid W_0, \mathcal{H}] = m_s n_t + \mathbf{1}\{j = k\} \text{Cov}(p_{s1}, q_{t1} \mid W_0, \mathcal{H}).$$

Summing coefficients and averaging proves (3). One Bernoulli(1/2) map pair, with all others constant, isolates $\Lambda_{st} = 1/16$ and proves necessity; sufficiency is immediate. The independent-copy covariance identity gives $\mathbb{E}[Q \mid \mathcal{H}] = \langle \Omega, \Lambda \rangle$. Centered comparisons lie in $[-1/2, 1/2]$, hence $|AB| \leq \|\mathbf{C}\|_1 \|\mathbf{D}\|_1 / 4$ and $|Q| \leq \|\Omega\|_1 / 2$. Hoeffding and empirical Bernstein each bound the same conditional mean; splitting their failure budget yields (4). The independent evaluation/validation allocation proves (5). Monotonicity in α proves the fixed- δ inversion.

For the comparator, average $(\sum_s c_s p_{s,ij})(\sum_t d_t q_{t,ik})$ over ordered distinct i, j, k . A scalar pair's six-role average lies in $[-1/12, 1/12]$: ordering the first map's values $x_1 < x_2 < x_3$ reduces 24 times this average to $-2 \text{sgn}(y_1 - y_3)$ for the second map's values; one tie gives a sum of two signs, and complete ties give zero. Triangle inequality gives W_U . Classical U-statistic concentration uses q disjoint triples. Since $q \geq \lfloor 2M/3 \rfloor + 1 > 2M/3$ and $\|c\|_1 \|d\|_1 \leq \|\mathbf{C}\|_1 \|\mathbf{D}\|_1$,

$$\frac{h(W_U, q, \alpha)}{h(W_S, M, \alpha - \delta)} \leq \frac{1}{3} \sqrt{\frac{M \log(1/\alpha)}{q \log\{1/(\alpha - \delta)\}}} < \frac{1}{\sqrt{6}}.$$

¹<https://github.com/MaxwellNi/certifying-forecasting-skill>.

If $W_S = 0$, both widths vanish and no radius ratio is asserted.

For the shared baseline, write $S = (x - z)(y - z)$ with uncentered comparisons $x, y, z \in \{0, 1/2, 1\}$. Conditioning on z gives $S \in [-1/4, 1]$. For $Q = (u - w)(v - w)/2$, all three differences lie in $[-1, 1]$; the product lies between $w^2 - 1$ and $(1 + |w|)^2$, giving $Q \in [-1/2, 2]$. Let $k(x, y)$ be the scalar six-role kernel above. Then $U = k(h, Y) - k(h, b) - k(b, Y) + k(b, b)$. A nonconstant triple of baseline values has $k(b, b) = 1/12$; using $|k| \leq 1/12$ gives $U \in [-1/6, 1/3]$. For a constant baseline, its three terms vanish, leaving $|U| \leq 1/12$. All three endpoint pairs are attained by three-point maps, including ties.

Proposition 5 applies the BY theorem [35], conditionally on selection when required, then averages. Marginal validity alone is insufficient: for uniform p and $\alpha < t$, $\Pr(p \leq \alpha \mid p \leq t) = \alpha/t$.

a) *Finite rank references:*

Proof of Proposition 2. For distinct i, j, l , conditional independence gives $\mathbb{E}[a_{ij}b_{il} \mid V_i, W_i, Z_i, H] = F_V(V_i)F_W(W_i)$. Averaging $(a_{ij} - f(Z_i))(b_{il} - g(Z_i))$ gives $\vartheta + b_H$; the k shared pairs in C_i add Γ/k .

Proof of Theorem 1. Symmetrize $(a_{12} - m_V(Z_1))(b_{13} - m_W(Z_1))$. The focal and reference role expectations are $r_V r_W, h_V(V), h_W(W)$, each integrating to ϑ , so the first projection is $\psi/3$. Bounded U-statistic decomposition [36] gives $D_j - \vartheta = N^{-1} \sum_i \psi(O_{ji}) + R_j$, $\mathbb{E}R_j = 0$, $\mathbb{E}R_j^2 = O(N^{-2})$. The scaled mean remainder has variance $O(N^{-1})$; the bounded-projection CLT applies for the fixed common law and positive limiting variance. Algebra gives $C_j - D_j = (J_j - Q_j)/(N - 1)$, where J_j, Q_j average shared pairs and distinct triples. Their difference has mean Γ and variance $O(N^{-1})$, so its centered scaled mean has variance $O(N^{-2})$. This proves both centered limits and $N \text{Var}(A_j) \rightarrow \sigma_\psi^2$. Disjoint-triple symmetrization gives $\mathbb{E}W_j^4 = O(N^{-2})$ for $W_j = A_j - \mathbb{E}A_j$. Thus $\text{Var}[(N/M) \sum_j W_j^2] = O(M^{-1})$ and $NW^2 = O_p(M^{-1})$ imply $N\hat{v}_A \rightarrow_p \sigma_\psi^2$. Slutsky gives the boundary limits and $T_C/[\Gamma\sqrt{MN}/\{(N - 1)\sigma_\psi\}] \rightarrow_p 1$ in the divergent case. Conditional independence centers the reference-role residuals. Boundedness makes the remainder and moment bounds uniform.

Proof of Corollary 1. Let $\delta_f = f - m_V$, $\delta_g = g - m_W$. The fitted-minus-oracle group difference, identical for C, D , is

$$\Delta_j = \frac{1}{N} \sum_i \{ -\delta_f(Z_i)(U_{W,i} - m_W(Z_i)) - \delta_g(Z_i)(U_{V,i} - m_V(Z_i)) + \delta_f(Z_i)\delta_g(Z_i) \}.$$

It is an order-two U-statistic with conditional mean b_H and, by bounded comparisons and clipped fits, $\text{Var}(\Delta_j \mid H) \leq c_0(\varepsilon_f^2 + \varepsilon_g^2)/N$ for an absolute constant c_0 . Conditional group independence gives $\sqrt{MN}(\bar{\Delta} - b_H) = o_p(1)$; the product-rate condition gives $\sqrt{MN}b_H = o_p(1)$. Conditional unbiasedness yields $N\hat{v}_\Delta = o_p(1)$, and sample covariance Cauchy-Schwarz

gives $N|\hat{v}_\Delta - \hat{v}_\Delta^o| \leq N\hat{v}_\Delta + 2\sqrt{N\hat{v}_\Delta N\hat{v}_\Delta^o} = o_p(1)$. The oracle limits and ratio argument transfer.

Proof of Proposition 3. Conditional on focal validation categories, rectangular pair comparisons have means $m_V(c), m_W(c)$; their intervals fail with probability at most $\delta/2$. The binomial limits, with total failure probability $\delta/2$, are $u_c = \text{Beta}^{-1}(1 - \delta/(2C_Z); n_c + 1, m_{\text{val}} - n_c)$ for $n_c < m_{\text{val}}$, and one otherwise. Since $u_c \geq n_c/m_{\text{val}}$, $\mathcal{P}$ is feasible. On the joint event it contains the true probability vector; bilinearity gives endpoint extrema and $\hat{B}_- \leq b_H \leq \hat{B}_+ \leq \hat{B}_{\text{abs}}$.

Here $\mathcal{H}_{\text{val}}$ denotes independent training and validation. Let σ^2, R be the disjoint-triple kernel variance and range. Permutation averaging and exponential convexity transfer the independent J -kernel Hoeffding and Bernstein moment bounds to $\bar{D}$ [24]. The upper-deviation radii $R\sqrt{x/(2J)}$ and $\sigma\sqrt{2x/J} + Rx/(3J)$ each fail with probability at most e^{-x} . Theorem 10 of [17] gives $\sigma \leq s + R\sqrt{2x/(J - 1)}$ except with probability e^{-x} . Conditional on $\mathcal{H}_{\text{val}}$, the smaller true-variance radius is fixed. Union with the variance event proves Eq. (13) at failure $2e^{-x}$, without independence of $s, \bar{D}$. Since $\mathbb{E}[\bar{D} \mid \mathcal{H}_{\text{val}}] = \vartheta + b_H$, adding validation failure δ proves the bound.

For fixed δ , the lower bound increases with α , so monotone inversion gives super-uniform p -values, including $p = 1$ when the corrected score does not exceed the learning allowance. Permutation averaging also transfers the negative-exponent Hoeffding bound. On the joint validation event, with evaluation probability at least $1 - \eta$, the lower certificate is at least $\vartheta - W - R[\sqrt{\log(1/(\alpha - \delta))} + \sqrt{\log(1/\eta)}]/\sqrt{2J}$, proving Eq. (15).

Proof of Lemma 1. If $\mathcal{S}$ is empty, the deterministic corner bound suffices. Otherwise, condition only on the focal validation categories, and write $a_c = m_V(c) - f(c)$, $b_c = m_W(c) - g(c)$ and $e_{Ac} = \hat{a}_c - a_c$, $e_{Bc} = \hat{b}_c - b_c$. The errors are independent across streams and categories, with sub-Gaussian proxies $1/(4n_{Ac})$ and $1/(4n_{Bc})$ by Hoeffding's lemma. Over $\mathcal{S}$,

$$\begin{aligned} \sum_c p_c(a_c b_c - \hat{a}_c \hat{b}_c) &= - \sum_c p_c \hat{b}_c e_{Ac} - \sum_c p_c \hat{a}_c e_{Bc} \\ &\quad + \sum_c p_c e_{Ac} e_{Bc}. \end{aligned}$$

Conditioning on the opposite stream gives upper-tail bounds r_A, r_B , each failing with probability at most e^{-x} . For independent centered sub-Gaussian X, Y with proxies s_X^2, s_Y^2 , condition on X and integrate a standard normal variable to obtain $\mathbb{E}e^{tXY} \leq (1 - t^2 s_X^2 s_Y^2)^{-1/2}$ for $|t|s_X s_Y < 1$. Thus $R_2 = \sum_c p_c e_{Ac} e_{Bc}$ obeys

$$\log \mathbb{E}e^{tR_2} \leq \frac{t^2 \sum_c d_c^2}{2(1 - t \max_c d_c)}, \quad 0 < t < 1/\max_c d_c.$$

Here $-\log(1 - u^2) \leq u^2/(1 - |u|)$. Chernoff optimization gives $\Pr(R_2 > r_{AB}) \leq e^{-x}$. A union bound over the three tails gives $3e^{-x} = \delta$ without requiring their events to be independent. Missing categories contribute at most their

deterministic corner bound; conditioning fixes the missing set. Remove conditioning to finish.

Proof of Proposition 4. The entity effect cancels: $y_r - \bar{y}_O = \varepsilon_r - \bar{\varepsilon}_O$. Shock orthogonality removes the other terms. For $r \in I$,

$$\frac{\mathbb{E}[(x_r - \bar{x}_O)(y_r - \bar{y}_O)]}{\sigma^2} = -\frac{\sum_{j \in O} w_{rj}}{m} - \frac{\sum_{s \in O} w_{sr}}{m} + \frac{\sum_{s,j \in O} w_{sj}}{m^2}.$$

The first two sums reduce to E and L , giving Eq. (17). In the split product, forecast shocks precede r and outcome shocks occur at or after r , so its expectation is zero. For nonlinear forecasts, set $\mathcal{G}_s = \sigma(\alpha, \eta_1, \dots, \eta_T, \varepsilon_1, \dots, \varepsilon_{s-1})$. For $s \geq r$, the forecast residual is $\mathcal{G}_s$ -measurable, giving $\mathbb{E}[(x_r - \bar{x}_E)\varepsilon_s] = 0$. Apply this at r and average over L .

Proof of Corollary 2. Each S_g has mean zero. Bounded fourth moments give Lyapunov ratio $O(G^{-1})$; independence gives $G^{-1} \sum_g \{S_g^2 - \mathbb{E}S_g^2\} \rightarrow_p 0$ and $\bar{S} \rightarrow_p 0$. The sample variance converges to v ; studentization completes the proof.

b) Fixed peers: Conditional on entity attributes, channel independence gives $\mathbb{E}[U_{V,i}U_{W,i}] = \bar{p}_i\bar{q}_i$ and $\mathbb{E}[a_{ij}b_{ij}] = p_{ij}q_{ij}$, where $\bar{p}_i = k^{-1} \sum_{j \neq i} p_{ij}$ and similarly for $\bar{q}_i$. Exact centering gives $\mathbb{E}C_i = 0$; Eq. (6) then gives Eq. (16).

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